

Advanced (Carolina Reaper)

- Q1.** A shopper moves between basement parking and upper retail levels. If their current level x satisfies $|x - 5| = 12$, which set represents the possible floors the shopper is on?
- (A) $\{-17, 7\}$
(B) $\{-7, 17\}$
(C) $\{7, 17\}$
(D) $\{-12, 12\}$
(E) $\{5, 12\}$
- Q2.** A hobbyist tracks the precision of a laser cutter. If the error e is defined as $2|s - 200| - |s - 100|$, evaluate the error when the setting $s = 85.5$.
- (A) 214.5
(B) 114.5
(C) 129.5
(D) 229.0
(E) 100.5
- Q3.** A chef notes that a refrigerator's temperature fluctuation T relative to 0°C satisfies $|T| \leq 3$. If the display shows d , and the error is $|d - T|$, what is the maximum value of $|d - T|$ if $d = 1$ and T is at its lower limit?
- (A) 1
(B) 2
(C) 3
(D) 4
(E) 0
- Q4.** In a marathon, two runners' positions on a straight track are $x = -|-4 - |-7||$ and $y = |-(-9)|$. Calculate the value of $|x \cdot y| - |x + y|$.
- (A) 101
(B) 89
(C) 97
(D) -97
- (E) 0
- Q5.** A luxury watch is tested for water resistance. If the depth d must satisfy $||d - 50| - 20| = 5$, what is the sum of all possible depths d that meet this criteria?
- (A) 200
(B) 100
(C) 150
(D) 250
(E) 180
- Q6.** An investor analyzes a balance sheet where $a < 0 < b$. Simplify the net variance expression: $|a| - |b - a| + |b|$.
- (A) $2b - 2a$
(B) $2b$
(C) $a + b$
(D) $-2a$
(E) 0
- Q7.** Two drones are at positions x and y on a 1D flight path. If $|x| = 15$ and $|y| = 22$, what is the minimum possible distance $|x - y|$ between them?
- (A) 37
(B) 15
(C) 7
(D) 0
(E) 22
- Q8.** A social media manager tracks follower growth. If the daily change c follows $c = |k - 10| - |k + 10|$, what is the value of c when $k > 10$?
- (A) 20
(B) -20
(C) $2k$
(D) $-2k$
(E) 0

Open Ended Questions (FRQ)

- Q9.** A scuba diver starts at -25 meters (below sea level). They descend 12 meters, then ascend 18 meters. Using absolute value notation, calculate the total vertical distance traveled and determine the final position d .
- Q10.** A performance bonus B is given by $B = 1000 - 5|s - 2000|$, where s is the total sales. If Salesperson A sells $s = 1800$ and Salesperson B sells $s = 2200$, use the definition of absolute value as distance to prove their bonuses are equal.

Chef's Tips

- Emphasize that absolute value measures distance, which can never be negative.
- Watch for students incorrectly distributing negative signs into the absolute value bars.
- Encourage drawing a number line for multi-layered absolute value equations.